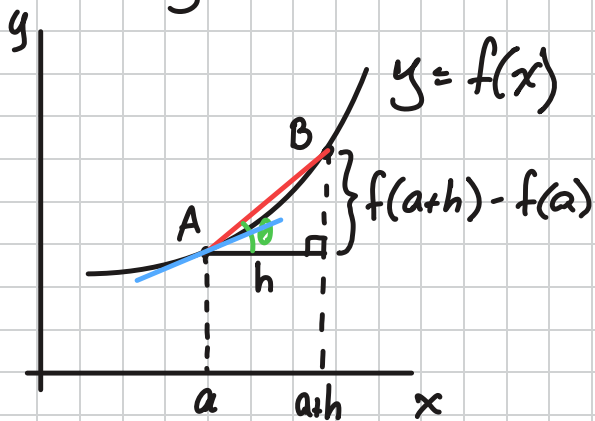


Week 4

derivatives

tangent lines



$$m = \tan \theta = \frac{f(a+h) - f(a)}{h} \quad \text{Slope of Secant AB}$$

to get tangent line at $x=a$
we take the limit as $h \rightarrow 0$

$$\text{Slope of tangent line: } m = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Example

$f(x) = \sqrt{x}$ Find eq of tangent line to $f(x)$ @ $a=4$

$$m = \lim_{h \rightarrow 0} \frac{f(4+h) - f(4)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{4+h} - 2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{4+h} - 2}{h} \times \frac{\sqrt{4+h} + 2}{\sqrt{4+h} + 2}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{4+h} - 4}{\cancel{h}(\sqrt{4+h} + 2)}$$

$$= \frac{1}{\sqrt{4+0} + 2} = \left(\frac{1}{4} \right)$$

now find point

$$y = \sqrt{x}$$

$$y = \sqrt{4}$$

$$= 2 \quad P(4, 2)$$

$\therefore$ at $P(4, 2)$ Slope is $\frac{1}{4}$

$$y - y_0 = m(x - x_0)$$

$$y - 2 = \frac{1}{4}(x - 4)$$

$$\Rightarrow y = \frac{x}{4} + 1$$

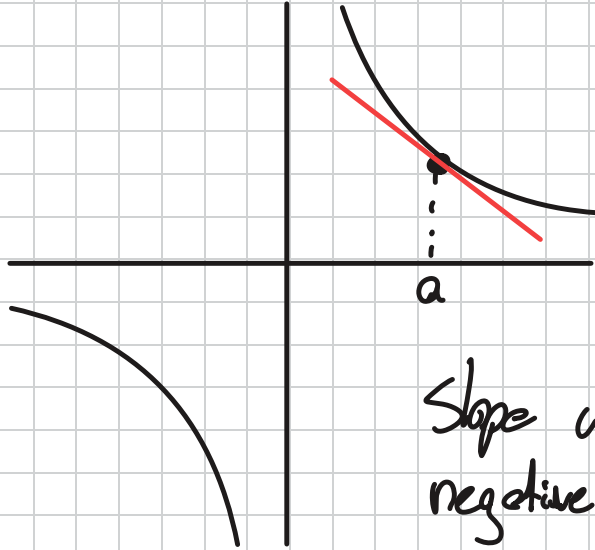
② find equation of normal line

$$m_n = -4$$

$$y - 2 = -4(x - 4)$$

$$y = -4x + 18$$

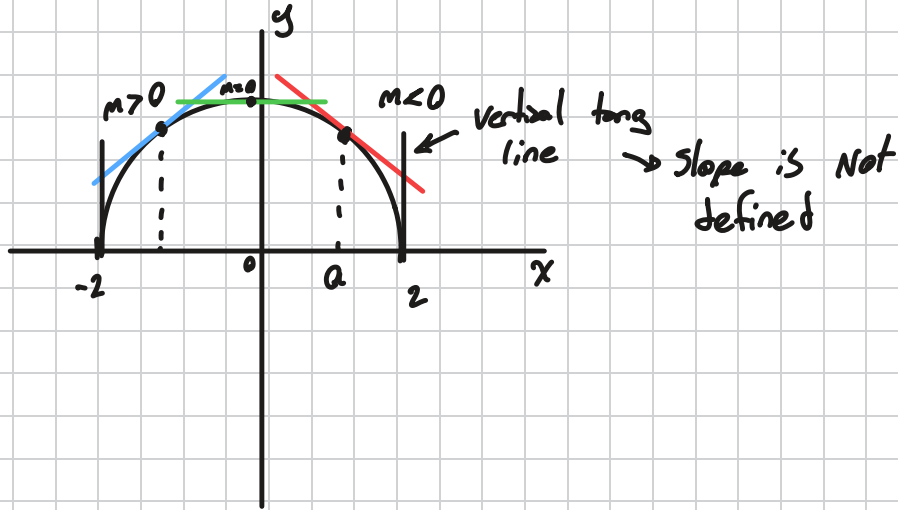
$$y = -\frac{1}{2}x + \frac{2}{a}$$



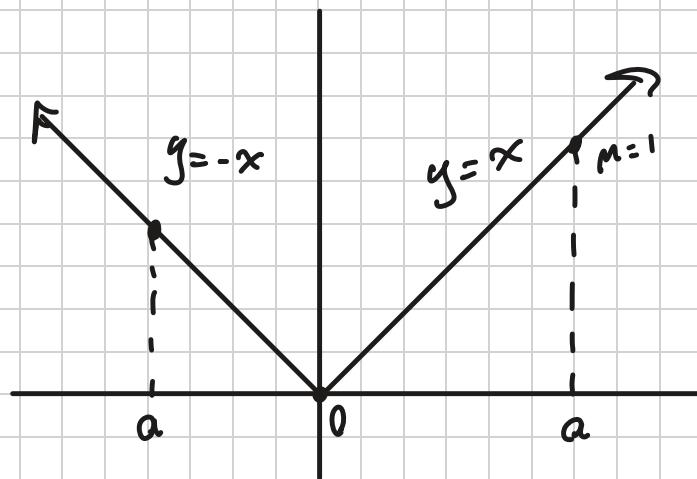
Slope will be
negative everywhere

Ex $f(x) = \sqrt{4-x^2}$ $4-x^2 \geq 0$ or $x^2 \leq 4$
 $\Rightarrow x \in [-2, 2]$

$y = \sqrt{4-x^2} \Rightarrow y^2 = 4-x^2$
 $\Rightarrow x^2 + y^2 = 4$ equation of Circle



Ex $f(x) = |x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$



No tang line
at $x=0$

Differentiable

$f(x)$ domain D

Def f is differentiable at $x \in D$
if $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ exists and is limit

then $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

the derivative of $f(x)$ at x

if f is differentiable at all $x \in D$

$\Rightarrow f'(x)$ is also a function on D

$$E_x \textcircled{1} \quad f(x) = x^2 \quad \rightarrow \quad f'(x) = 2x$$

$$D(f) = \mathbb{R} \quad \rightarrow \quad D(f') = \mathbb{R}$$

$$\textcircled{2} \quad f(x) = \frac{1}{x} \quad \downarrow \quad D(f) = \mathbb{R} - \{0\}$$

$$f'(x) = -\frac{1}{x^2} \quad \downarrow \quad D(f') = \mathbb{R} - \{0\}$$

$$y = \frac{1}{x}$$

$$y' = -\frac{1}{x^2} = -2$$

$$m = -2$$

$$x_0 = -\frac{\sqrt{2}}{2}, \quad y_0 = -\sqrt{2}$$

$$\Rightarrow y - y_0 = m(x - x_0)$$

$$y + \sqrt{2} = -2\left(x + \frac{\sqrt{2}}{2}\right) = -2x - \sqrt{2}$$

$$\Rightarrow \boxed{y = -2x - \sqrt{2}} \quad \text{through } \left(-\frac{\sqrt{2}}{2}, -\sqrt{2}\right)$$

Exercise: repeat for $x_0 = \frac{\sqrt{2}}{2}$

Differentiation rules

$$y = f(x)$$

$$\frac{d}{dx} \left(\frac{1}{f(x)} \right) = - \frac{f'(x)}{(f(x))^2} \quad \text{reciprocal rule}$$

Ex $g(x) = 2x + 7$ find $\frac{d}{dx} \left(\frac{1}{2x+7} \right)$

$$\frac{d}{dx} \left(\frac{1}{2x+7} \right) = - \frac{2}{(2x+7)^2}$$

the Quotient rule

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$$

I remember
d3 Shi

Product rule
(1 remember)

$$\frac{d}{dx} \left(\frac{ax+b}{cx+d} \right) = \frac{a(cx+d) - c(ax+b)}{(cx+d)^2}$$
$$= \frac{ad - bc}{(cx+d)^2}$$